

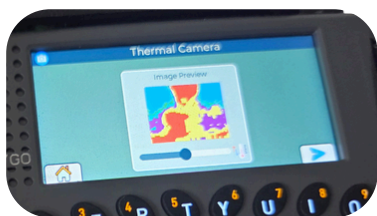


Adding a Thermal Camera to Lilygo Pager

The Blackout Comms off-grid messaging & location firmware is capable of working with several add-on hardware components. One of those is the Adafruit MLX90640 thermal camera.

This thermal camera is fun to play with, as well and being useful for finding hot spots and cold spots very easily (up close).

Just aware the camera can't see too far and the resolution & frame rate are not very high. If that doesn't matter to you, give it a go!

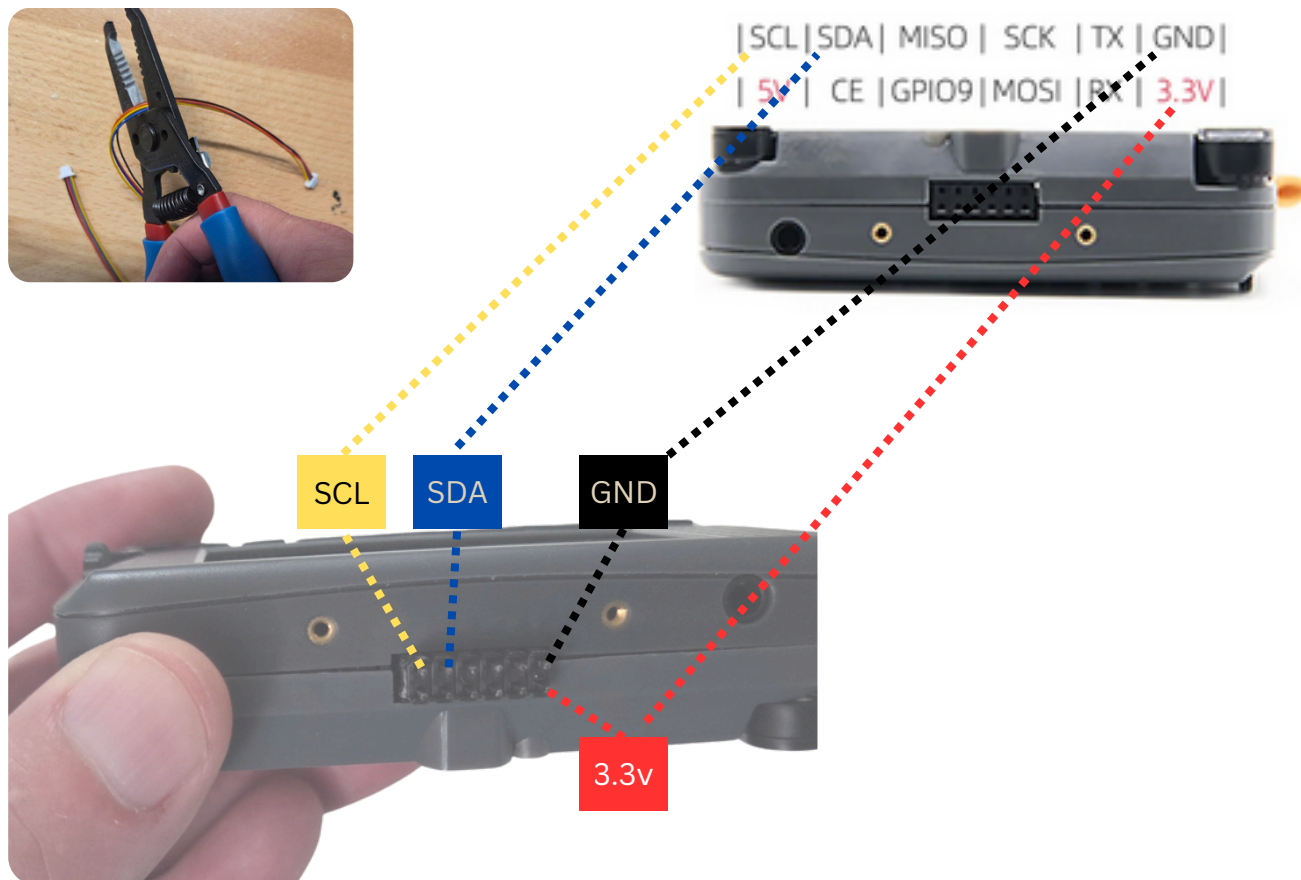
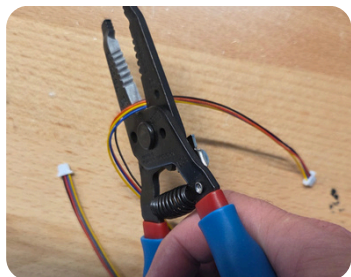


FIRST: Install Blackout Comms

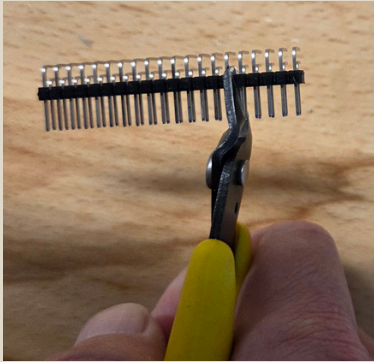
Download Blackout Comms for Lilygo Pager are at:
<https://www.chatters.io/pager>

Pager Expansion Port & Header Pins

You will need to create an adapter between the pager expansion port and a standard QWIIIC cable. Once this is done, the pager is ready to take Blackout Comm compatible modules.



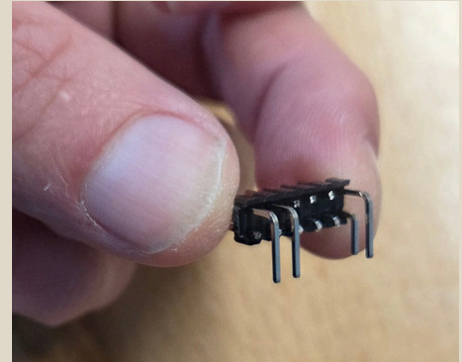
Prepare a set of header pins that match the Pager



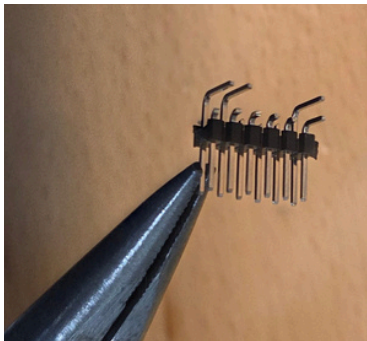
Using a 90° double-row pin header, trim it to 6x2.

Trim all the unused pins so they're not in the way.

We only need SDA, SCL, GND and 3.3v.



Solder the QWIIC Wires to the Header Pins



Use heat shrink insulator and/or liquid electrical tape to insulate the pins and bare wire, so they can't come in contact with one another.

Plug the Adapter into the Pager



The wires should be directed toward the bottom of the pager, pins facing down, away from the pager's screen.



Connect the Camera & Power on the Pager Momentarily



If the camera gets hot and/or the light doesn't come on, unplug it immediately and double-check the wiring!

- The camera should light up green
- The home screen should display a camera icon

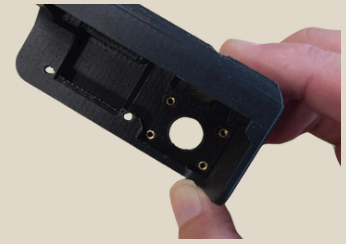
Power it back off (tap middle button 2x)

Install the 4 Knurled Nuts



The printed backpack/housing has 4 holes for heat sinking M2 knurled nuts into.

The easiest way to sink these is with a soldering tip for melting knurled nuts into place.



Mount the Camera Into Place



The camera should fit just right as shown, and you can use very short M2 screws to mount it.

As you place the backpack over the pager, ensure no wires are pinched/trapped.

Install the Backpack Onto the Pager



Use 4 M2 screws to attach the backpack onto the pager. Now you should be ready to power it on and see it work!

Parts List

To assemble using the instructions above, you'll need soldering equipment & supplies, as well as the components listed below:



Lilygo Pager

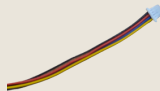
[Rokland](#), [Lilygo](#)

Only compatible with SX1262 Models



MLX90640 Thermal Camera

[Adafruit](#)



QWIIC Cable

[Amazon](#)



M2 Knurled Nuts

[Amazon](#)



M2 Screws

[Amazon](#)



90 Degree 2 Row Header Pins

[Amazon](#)